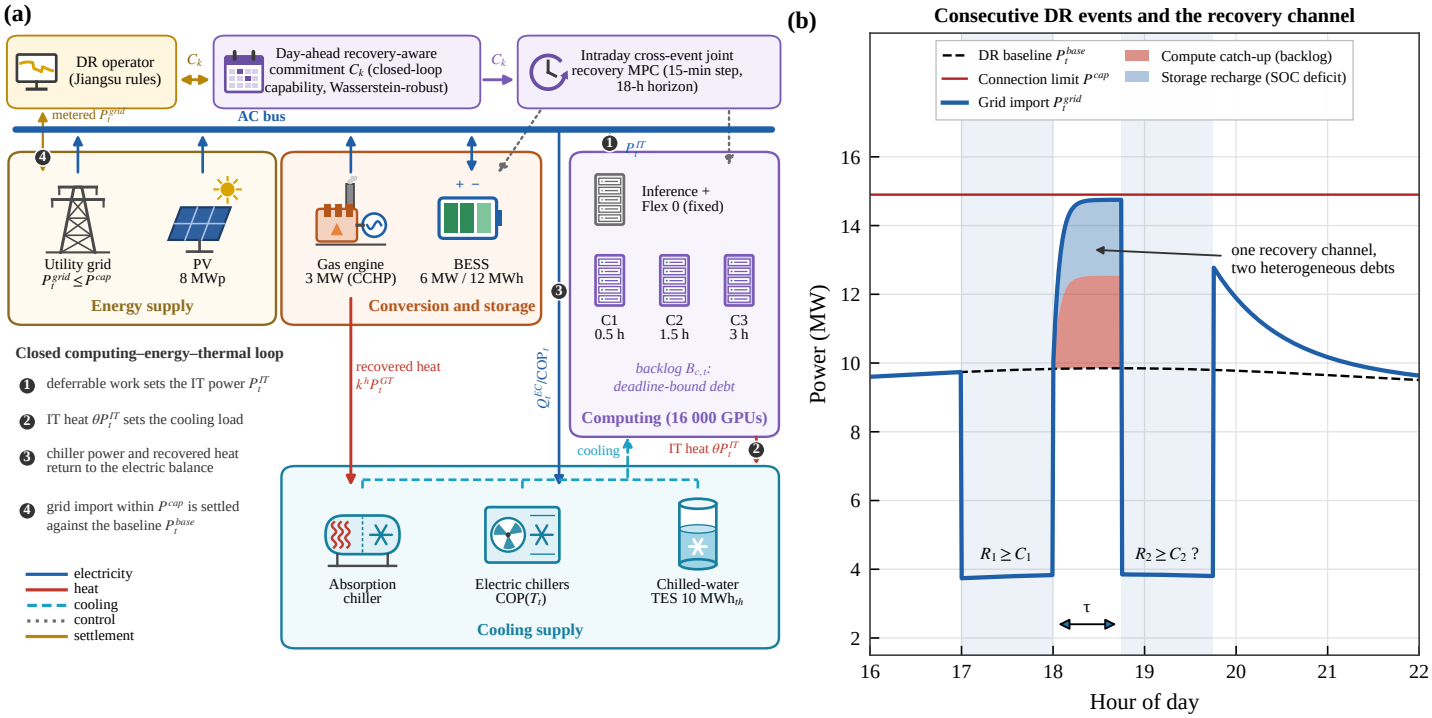

(a)
DR operator (Jiangsu rules)
C_k
Day-ahead recovery-aware commitment C_k (closed-loop capability, Wasserstein-robust)
C_k
Intraday cross-event joint recovery MPC (15-min step, 18-h horizon)
AC bus
metered P_t^{grid}
Utility grid $P_t^{grid} \le P^{cap}$
PV 8 MWp
Energy supply
Gas engine 3 MW (CCHP)
BESS 6 MW / 12 MWh
Conversion and storage
P_t^{IT}
Inference + Flex 0 (fixed)
C1 0.5 h
C2 1.5 h
C3 3 h
backlog $B_{c,t}$: deadline-bound debt
Computing (16 000 GPUs)
recovered heat $k^h P_t^{GT}$
Q_t^{EC}/COP_t
cooling
IT heat θP_t^{IT}
Absorption chiller
Electric chillers COP(T_t)
Chilled-water TES 10 MWh$_{th}$
Cooling supply
Closed computing–energy–thermal loop
1 deferrable work sets the IT power P_t^{IT}
2 IT heat θP_t^{IT} sets the cooling load
3 chiller power and recovered heat return to the electric balance
4 grid import within P^{cap} is settled against the baseline P_t^{base}
electricity
heat
cooling
control
settlement
(b)
Consecutive DR events and the recovery channel
DR baseline P_t^{base}
Connection limit P^{cap}
Grid import P_t^{grid}
Compute catch-up (backlog)
Storage recharge (SOC deficit)
one recovery channel, two heterogeneous debts
$R_1 \ge C_1$
$R_2 \ge C_2$?
τ
Power (MW)
Hour of day
16
14
12
10
8
6
4
2
16
17
18
19
20
21
22